HKU BUSINESS SCHOOL
港大經管學院

IIMT3642: Managing and Mining Big Data - Data Preparation

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Innovation and Information Management

Outline

- Data Concepts and Challenge
  - Attributes and Objects
  - Types of Dataset
  - Data Quality
- Data Preprocessing
  - Cleaning
  - Integration
  - Transformation
  - Reduction

What Is Data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

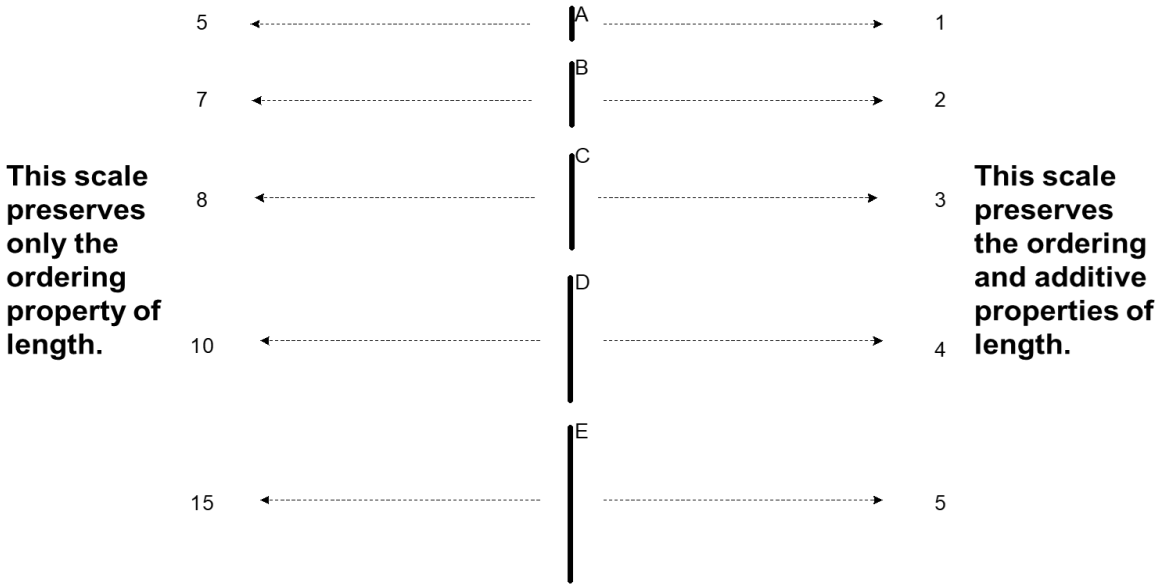
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute vs. Attribute Values

- Attribute values are numbers or symbols assigned to an attribute for a particular object
- Distinction between attributes and attribute values
  - Same attribute can be mapped to different attribute values
    - Example: height can be measured in feet or meters
  - Different attributes can be mapped to the same set of values
    - Example: Attribute values for ID and age are integers
    - But properties of attribute values can be different

Value the Attribute

For example, the measurement of length



- The way you measure an attribute may not match its properties

Types of Attributes

- There are different types of attributes
  - Nominal
    - Examples: ID numbers, eye color, zip codes
  - Ordinal
    - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
  - Interval
    - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
  - Ratio
    - Examples: temperature in Kelvin, length, time, counts

Properties of Attribute Values

- The type of an attribute depends on which of the following properties it possesses:
  - Distinctness: $= \neq$
  - Order: $< >$
  - Addition: $+ -$
  - Multiplication: $* /$

- Nominal attribute: distinctness
- Ordinal attribute: distinctness & order
- Interval attribute: distinctness, order & addition
- Ratio attribute: all 4 properties above

Difference Between Ratio and Interval

- Is it physically meaningful to say that a temperature of 10° degrees twice that of 5° on
  - the Celsius scale? - N
  - the Fahrenheit scale? - N
  - the Kelvin scale? - Y
- Consider an attribute measuring the <u>height above average</u>
  - If Bill's height is three inches above average and Bob's height is six inches above average, then would we say that Bob is twice as tall as Bob?
  - Is this situation analogous to that of temperature?

	Attribute Type	Description	Examples	Operations
Categorical Qualitative	Nominal	Nominal attribute values only distinguish. (=, ≠)	zip codes, employee ID numbers, eye color, sex: {*male, female*}	mode, entropy, contingency correlation, $\chi 2$ test
	Ordinal	Ordinal attribute values also order objects. (<, >)	hardness of minerals, {*good, better, best*}, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Numeric Quantitative	Interval	For interval attributes, differences between values are meaningful. (+, -)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, *t* and *F* tests
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	temperature in Kelvin, monetary quantities, counts, age, mass, length, current	geometric mean, harmonic mean, percent variation

This categorization of attributes is owing to S. S. Stevens

	Attribute Type	Transformation	Comments
Categorical Qualitative	Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
	Ordinal	An order preserving change of values, i.e., *new_value = f(old_value)* where *f* is a monotonic function	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by { 0.5, 1, 0}.
Numeric Quantitative	Interval	*new_value =a * old_value + b* where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
	Ratio	*new_value = a * old_value*	Length can be measured in meters or feet.

Discrete and Continuous Attributes

- Discrete Attribute
  - Has only a finite or countably infinite set of values
  - Examples: zip codes, counts, or the set of words in a collection of documents
  - Often represented as integer variables.
  - Note: binary attributes are a special case of discrete attributes
- Continuous Attribute
  - Has real numbers as attribute values
  - Examples: temperature, height, or weight.
  - Practically, real values can only be measured and represented using a finite number of digits.
  - Continuous attributes are typically represented as floating-point variables.

Asymmetric Attributes

- Presence of certain values is regarded as NOT equally important
- Examples:
  - Words present in documents (A list of dictionary words)
  - Items present in customer transactions (A basket of catalog products)
- If we met a friend in the grocery store would we ever say the following?

"I see our purchases/tastes are very similar since we didn't buy most of the same things."

- We may use <u>two asymmetric binary attributes</u> to represent <u>one ordinary binary attribute</u>
- For example, *purchased_no* in some recommendation scenarios are as informative as *purchased_yes*

Cross-sale

					Purchased (0/1)	
User_ID	**Browser History**	**Past subscriptions**	**Search Keywords**	**Dwelling Time**	**Purchased_No**	**Purchased_Yes**
1	…	…	…	…	0	1
2	…	…	…	…	1	0

Promotional

Types of Datasets

- Record
  - Data Matrix
  - Document Data
  - Transaction Data
- Graph
  - World Wide Web
  - Traffic/Social Networks
  - Molecular Structures
- Ordered
  - Spatial Data
  - Temporal Data
  - Sequential Data
  - Genetic Sequence Data

Important Characteristics of Structured Data

- Dimensionality
  - Curse of Dimensionality
- Sparsity
  - Only presence counts
- Resolution
  - Patterns depend on the scale

Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

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Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an *m* by *n* matrix, where there are *m* rows, one for each object, and *n* columns, one for each attribute

Example: materials' response to applied load

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document Data

- Each document becomes a ‘term’ vector,
  - each term is a component (attribute) of the vector,
  - the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

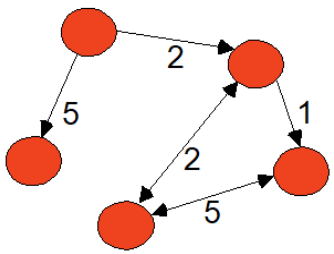
Transaction Data

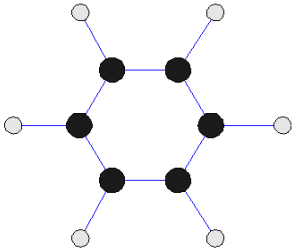
- A special type of record data, where
  - each record (transaction) involves a set of items.
  - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

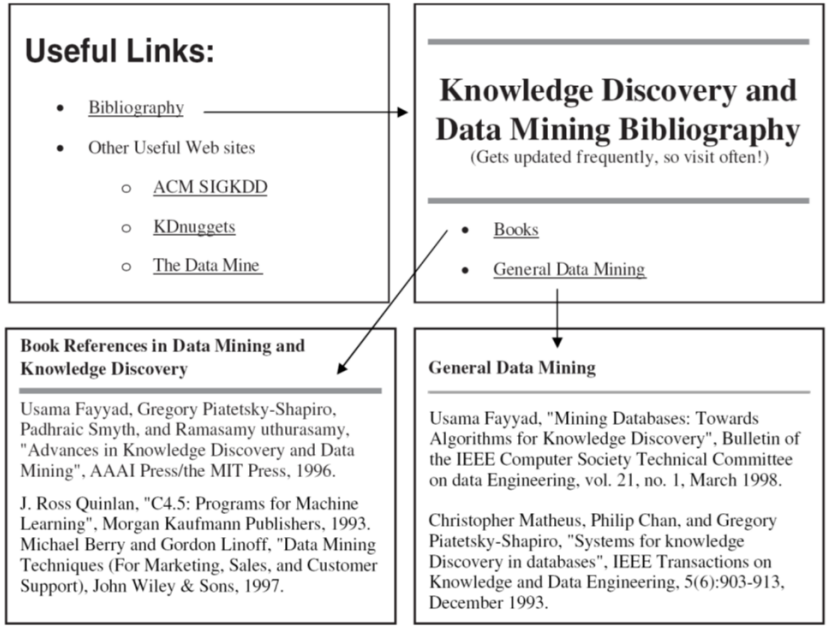
Graph Data

- Examples: Friendship Links, a Molecule Structure, and interlinked Webpages





Benzene Molecule: C6H6



Ordered Data

- Sequences of transactions

Items/Events

(A B) (D) (C E)
(B D) (C) (E)
(C D) (B) (A E)

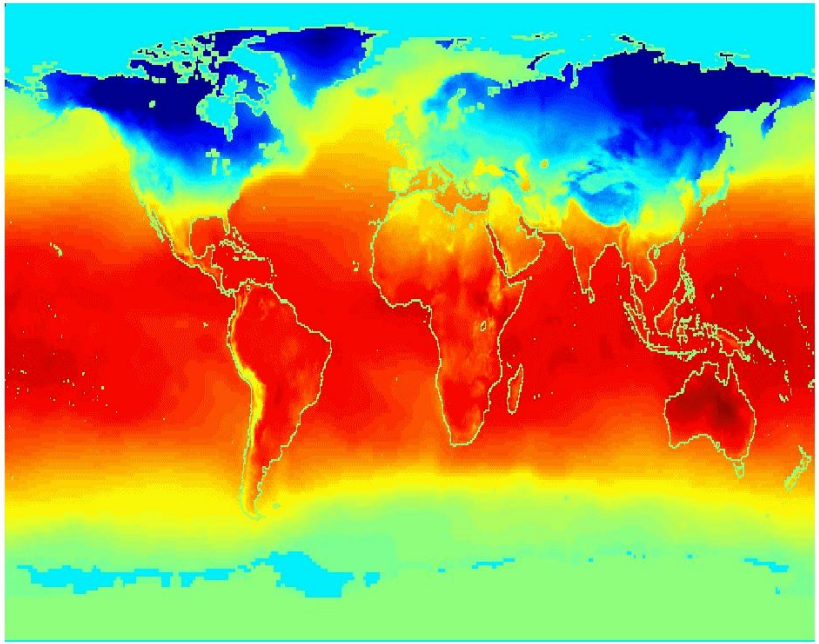
An element of the sequence

- Genomic sequence data (ACGT)

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

- Spatio-Temporal Data

Jan



Average Monthly Temperature of land and ocean

Data Quality, as a critical issue

- Poor data quality negatively affects many data processing efforts

  "The most important point is that poor data quality is an unfolding disaster. Poor data quality costs the typical company at least ten percent (10%) of revenue; twenty percent (20%) is probably a better estimate."

  —Thomas C. Redman, DM Review, August 2004

- A general principle applies in DM/ML: Garbage in, garbage out

In practice, we would like to know

- What kinds of data quality problems matter?
- How can we detect problems with the data?
- What can we do about these problems?

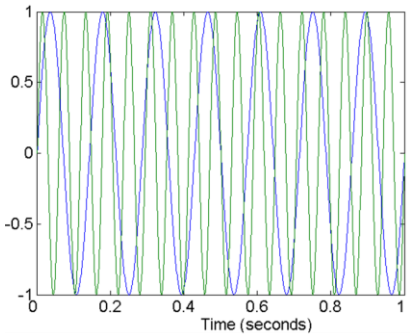
In practice, we would like to know

- **What kinds of data quality problems matter?**
- How can we detect problems with the data?
- What can we do about these problems?

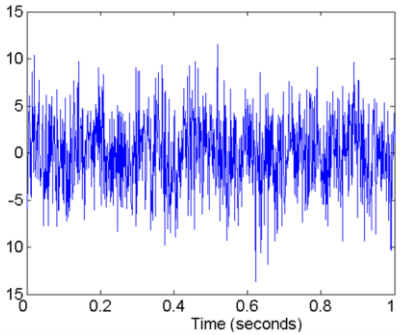
- Typical data quality issues:
  - Noise and outliers
  - Missing values
  - Duplicates/Inconsistencies

Noise

- Noise refers to (undesired) modification of original values, such as random error or variance in a measured variable
  - Examples: distortion of a person's voice when talking on a poor phone and "snow" on television screen



Two Sine Waves

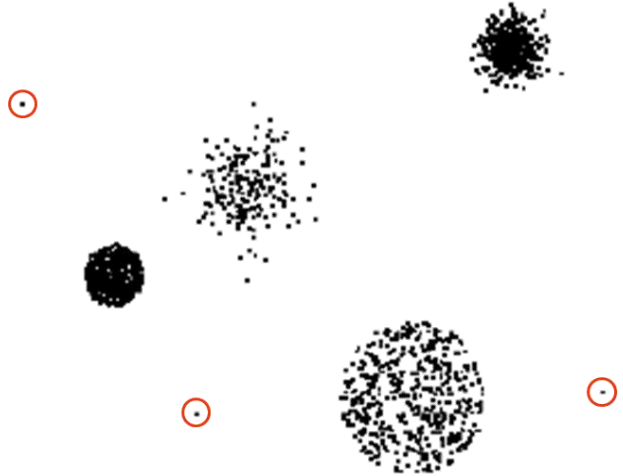


Two Sine Waves + Noise

  - Sources: measurement errors, input mistakes, transmission issues, environmental influences

Outliers

- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set
  - Case 1: Outliers are considered noise that interferes with data analysis
  - Case 2: Outliers are the goal of our analysis (Anomaly Detection)
    - Credit card fraud
    - Intrusion detection



- **<u>Discussion:</u>** outlier vs. noise?

Missing Values

- Reasons for missing values
  - Information is not collected
    (e.g., people decline to give their age and weight)
  - Attributes may not be applicable to all cases
    (e.g., annual income is not applicable to children)
- Handling missing values
  - Eliminate data objects
  - Estimate missing values
    - Example: time series of temperature
    - Example: census results
  - Ignore the missing value during specific analyses

Duplicate Data

- Data set may include data objects that are duplicates, or almost duplicates of one another
  - Major issue when merging data from heterogeneous sources
- Examples:
  - Same person with multiple email addresses
- Data cleaning
  - Process of dealing with duplicate data issues

Data Preprocessing

Data preprocessing involves multiple processes:

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction

1. **Data Cleaning**
2. Data Integration
3. Data Transformation
4. Data Reduction

- Missing values
- Noises
- Others

Data Cleaning

- Missing values
  - Deletion
    - Listwise deletion (remove entire row with missing values. Useful when the dataset is large and the missing values are few)
    - Pairwise deletion (only remove rows with missing values for specific analyses, e.g., cosine similarity)
  - Imputation
    - Univariate Imputation (Mean/Median/Mode imputation)
    - Multivariate Imputation (K-nearest neighbors imputation/Regression imputation*)
    - Iterative Imputation
  - Indicator Method
- <u>Hands-on Examples</u>

Data Cleaning

- Noises
  - Binning + Smoothing
  - Regression
  - Outlier

Data Cleaning

- Noise
  - **Binning + Smoothing**
  - Regression
  - Outliers

- Binning (dividing the range of the data into a series of intervals, aka. bins)
- Sorting (group the values by bin)
- Smoothing (averaging values with a sliding window)
  - Moving Average (MA)
  - Exponential Moving Average (EMA, more weights to recent data points)

Hands-on Example

Data Cleaning

- Noise
  - Binning + Smoothing
  - **Regression**
  - Outliers

➢ Handling
  - Smooth by fitting the data into regression functions
  - Same as we treat missing values

<u>Hands-on Example</u>

Data Cleaning

- Noise
  - Binning + Smoothing
  - Regression
  - **Outliers**

➢ Detection
  - Z-score
  - Interquartile Range (IQR)/Boxplot
  - Clustering

➢ Handling
  - Single outlier values are treated as missing data
  - Identified outlier data records can be removed

Hands-on Example

Data Cleaning

- Other Issues
  - Duplicate records
    - Consistent ID or Name (Remove redundant records)
    - Inconsistent ID or Name (Entity Identification)
      - E.g., Bill Clinton = William Clinton
  - Value conflicts
      - E.g., different scales like metric vs. British units

1. Data Cleaning
2. **Data Integration**
3. Data Transformation
4. Data Reduction

- Enrich the current data by combining from multiple sources
- Remove redundancies if any
  - Records (entity recognition)
  - Columns (correlation analysis)

1. Data Cleaning
2. Data Integration
3. **Data Transformation**
4. Data Reduction

A function applies to map the entire set of values of a given attribute into a new set of replacement values

- Smoothing (for data noises, **covered**)
- Attribute construction (create new attributes from existing ones)
- Normalization (scale to fall within a specified range)
- Discretization (convert a continuous attribute into an ordinal attribute)
- One-hot Encoding

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A function applies to map the entire set of values of a given attribute into a new set of replacement values

- Smoothing (for data noises, **covered**)
- Attribute construction (create new attributes from existing ones)
- Aggregation (combining two or more attributes (or objects) into a single attribute (or object))
- Normalization (scale to fall within a specified range)
  - Min-max normalization (linearly fall in [0,1])
  - Z-score normalization (may result in negatives)
  - **Hands-on Example**
- Discretization (convert a continuous attribute into an ordinal attribute)
- One-hot Encoding

1. Data Cleaning
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A function applies to map the entire set of values of a given attribute into a new set of replacement values

- Smoothing (for data noises, **covered**)
- Attribute construction (create new attributes from existing ones)
- Normalization (scale to fall within a specified range)
- Discretization (convert a continuous attribute into an ordinal attribute)
  - Equal-width binning
  - Equal-frequency binning
  - **Hands-on Example**
- One-hot Encoding (convert categorical data into a vectorized representation of 0s and 1)
  - **Hands-on Example**

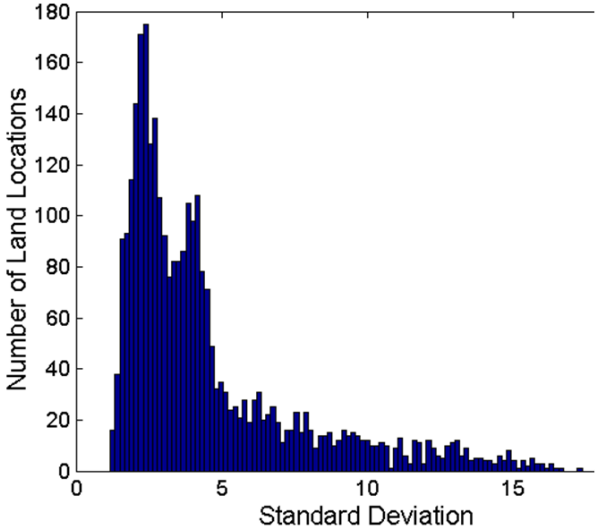
1. Data Cleaning
2. Data Integration
3. Data Transformation
4. **Data Reduction**

- Aggregation (combining two or more attributes (or objects) into a single attribute (or object))
- Sampling
- Dimensionality reduction

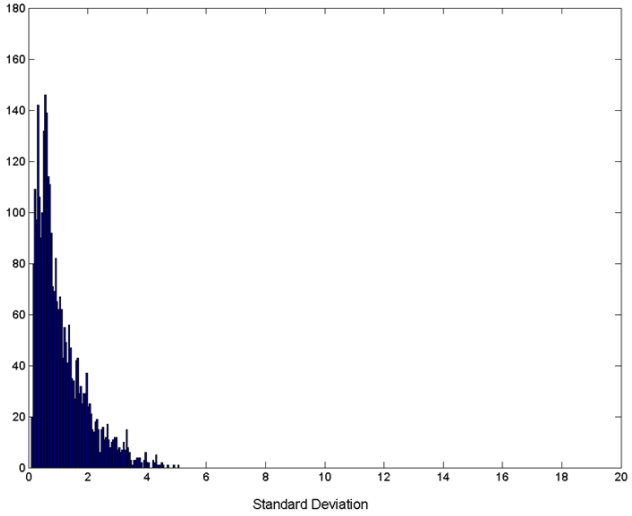
Aggregation Example: Precipitation in Australia

- This example is based on precipitation in Australia from the period 1982 to 1993. The next slide will show
  - A histogram for the standard deviation of average monthly precipitation for 3,030 0.5◦ by 0.5◦ grid cells in Australia, and
  - A histogram for the standard deviation of the average yearly precipitation for the same locations.
- The average yearly precipitation has less variability than the average monthly precipitation.
- All precipitation measurements (and their standard deviations) are in centimeters.

Variation of Precipitation in Australia



Standard Deviation of Average Monthly Precipitation



Standard Deviation of Average Yearly Precipitation

1. Data Cleaning
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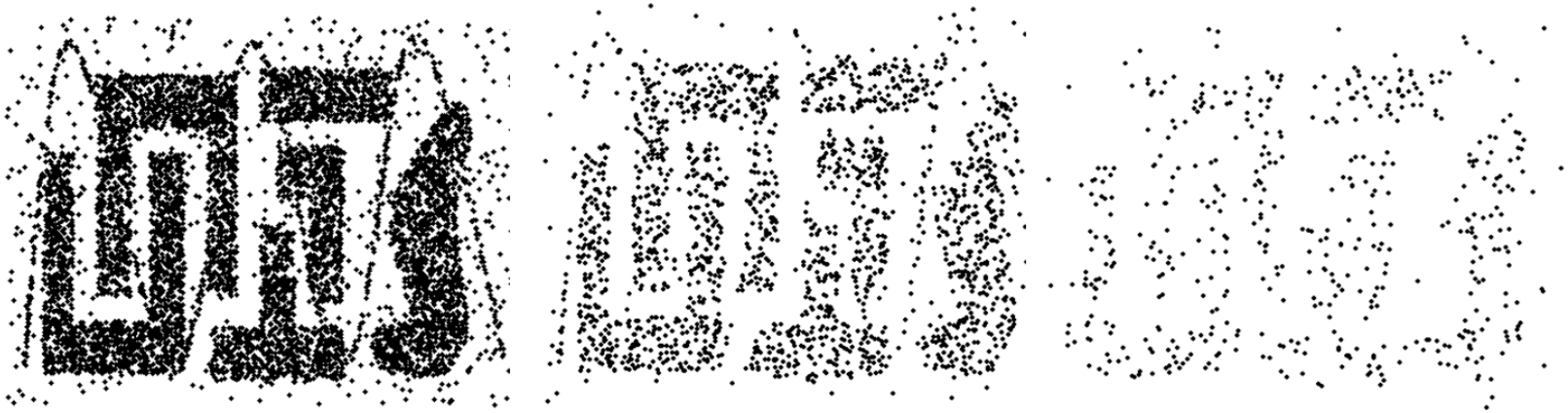
4. **Data Reduction**

- Aggregation (combining two or more attributes (or objects) into a single attribute (or object))
- Sampling
- Dimensionality reduction

Sampling

- The key principle for effective sampling is the following:
  - Using a sample will work almost as well as using the entire data sets, <u>if the sample is representative</u>
  - A sample is representative if it has approximately the same property (of interest) as the original set of data

Sample Size



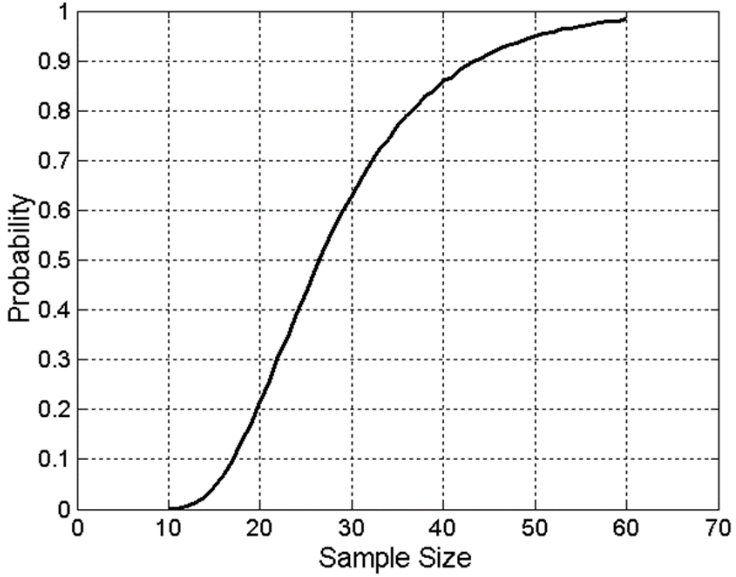
8000 points

2000 Points

500 Points

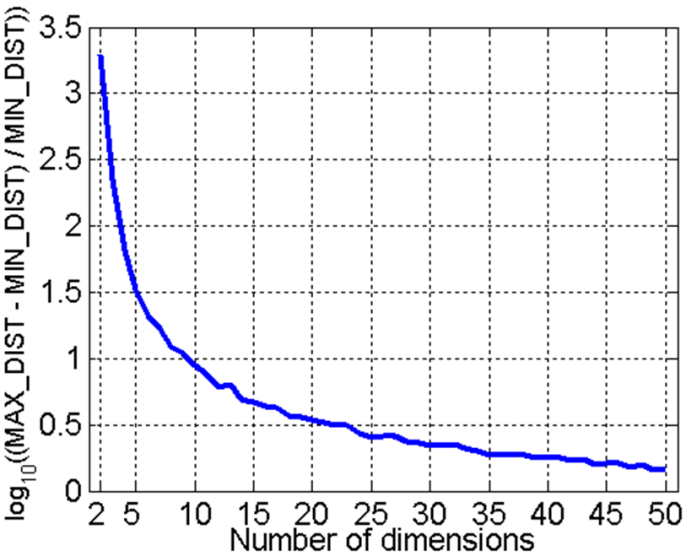
Sample Size

- What sample size is necessary to get at least one object from each of 10 equal-sized groups.



Curse of Dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies
- Definitions of density and distance between points, which is critical for clustering and outlier detection, become less meaningful



- **Randomly generate 500 points**
- **Compute difference between max and min distance between any pair of points**

1. Data Cleaning
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- Aggregation (combining two or more attributes (or objects) into a single attribute (or object))
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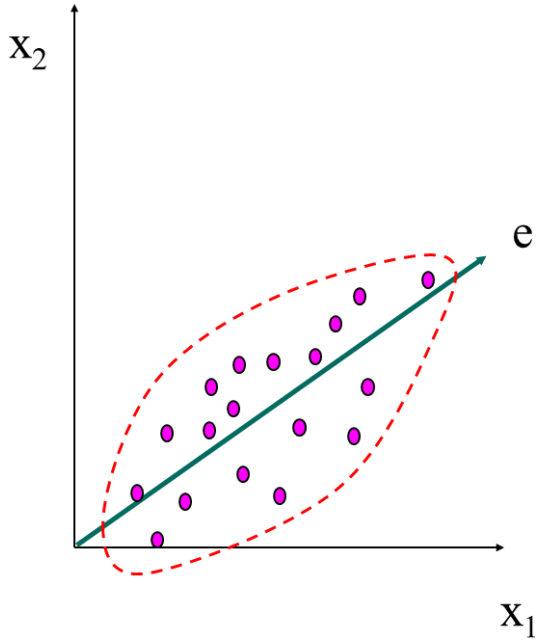
Dimensionality Reduction

- Purpose:
  - Avoid curse of dimensionality
  - Reduce amount of time and memory required by data mining algorithms
  - Allow data to be more easily visualized
  - May help to eliminate irrelevant features or reduce noise
- Techniques
  - Principal Components Analysis (PCA)
  - Singular Value Decomposition
  - Others: supervised and non-linear techniques

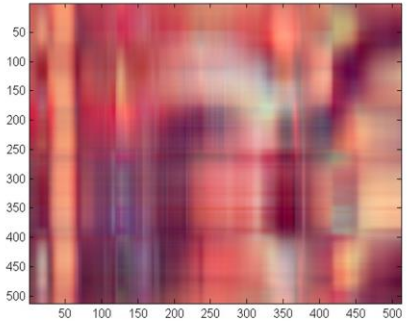
Dimensionality Reduction: PCA

- Goal is to find a projection that captures the largest amount of variation in data

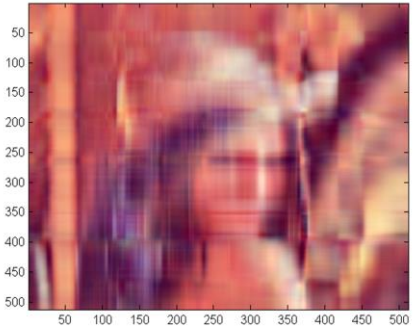


Top-N components with N = [4, 8, 16, 32, 256]

4



8



16

32

256

Feature Subset Selection

This is another way to reduce dimensionality of data:

- Redundant features (correlation analysis)
  - Duplicate much or all of the information contained in one or more other attributes
  - Example: purchase price of a product and the amount of sales tax paid
- Irrelevant features
  - Contain no information that is useful for the data mining task at hand
  - Example: students' ID is often irrelevant to the task of predicting students' GPA